HOMEWORK-3:

1.Dynamic allocation_store n elements and first largest

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i, largest;
    int *a;

    printf("Enter n: ");
    scanf("%d", &n);

    a = (int *)malloc(n * sizeof(int));

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    largest = a[0];

    for(i = 1; i < n; i++) {
        if(a[i] > largest)
            largest = a[i];
    }

    printf("Largest = %d", largest);

    free(a);
    return 0;
}
```

OUTUT:

```
Enter n: 4
Enter elements:
23
67
9
0
Largest = 67
```

2.Print array in reverse order

```
#include <stdio.h>

int main() {
    int a[100], n, i;
    int *p;

    printf("Enter n: ");
    scanf("%d", &n);
```

```
    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    p = &a[n - 1];

    printf("Reverse order:\n");
    for(i = n - 1; i >= 0; i--)
        printf("%d ", *p--);

    return 0;
}
```

```
OUTPUT:
Enter n: 6
Enter elements:
23
4
5
67
89
0
Reverse order:
0 89 67 5 4 23
```

---

3. Student information using structure and sort by name

```
#include <stdio.h>
#include <string.h>

struct Student {
    int roll;
    char name[50];
    float mark;
};

int main() {
    struct Student s[50], temp;
    int n, i, j;

    printf("Enter number of students: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("Enter roll, name and mark:\n");
        scanf("%d %s %f", &s[i].roll, s[i].name, &s[i].mark);
    }

    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(strcmp(s[i].name, s[j].name) > 0) {
```

```
                temp = s[i];
                s[i] = s[j];
                s[j] = temp;
            }
        }
    }

    printf("\nStudents sorted by name:\n");

    for(i = 0; i < n; i++)
        printf("%d %s %.2f\n",
               s[i].roll, s[i].name, s[i].mark);

    return 0;
}
```

```
OUTPUT:
Enter number of students: 3
Enter roll, name and mark:
15
Rahul
90
Enter roll, name and mark:
23
Pavi
88
Enter roll, name and mark:
35
Harish
91

Students sorted by name:
35 Harish 91.00
23 Pavi 88.00
15 Rahul 90.00
--------------------------------------------------------------------------------
--------
```

4.Add n distances using structure

```
#include <stdio.h>

struct Distance {
    int feet;
    float inch;
};

int main() {
    struct Distance d[50], sum = {0, 0};
    int n, i;

    printf("Enter number of distances: ");
    scanf("%d", &n);
```

```
    for(i = 0; i < n; i++) {
        printf("Enter feet and inches: ");
        scanf("%d %f", &d[i].feet, &d[i].inch);

        sum.feet += d[i].feet;
        sum.inch += d[i].inch;
    }

    sum.feet += (int)(sum.inch / 12);
    sum.inch = (int)sum.inch % 12;

    printf("Total distance = %d feet %.2f inches",
           sum.feet, sum.inch);

    return 0;
}
OUTPUT:
Enter number of distances: 3
Enter feet and inches: 12
0.4
Enter feet and inches: 23
9
Enter feet and inches: 12
5.3
Total distance = 48 feet 2.00 inches
--------------------------------------------------------------------------------
--------
5.Add two complex numbers by passing structure to function
#include <stdio.h>

struct Complex {
    float real;
    float imag;
};

struct Complex add(struct Complex c1, struct Complex c2) {
    struct Complex result;

    result.real = c1.real + c2.real;
    result.imag = c1.imag + c2.imag;

    return result;
}

int main() {
    struct Complex c1, c2, result;

    printf("Enter first complex number: ");
    scanf("%f %f", &c1.real, &c1.imag);
```

```
    printf("Enter second complex number: ");
    scanf("%f %f", &c2.real, &c2.imag);

    result = add(c1, c2);

    printf("Sum = %.2f + %.2fi",
           result.real, result.imag);

    return 0;
}
OUTPUT:
Enter first complex number: 12
8
Enter second complex number: 4
7
Sum = 16.00 + 15.00i
--------------------------------------------------------------------------------
--------
6. Inventory Management System using Structure
#include <stdio.h>

struct Item {
    int id;
    char name[50];
    int quantity;
    float price;
};

int main() {
    struct Item item[50];
    int n, i;

    printf("Enter number of items: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("\nEnter item ID: ");
        scanf("%d", &item[i].id);

        printf("Enter item name: ");
        scanf("%s", item[i].name);

        printf("Enter quantity: ");
        scanf("%d", &item[i].quantity);

        printf("Enter price: ");
        scanf("%f", &item[i].price);
    }
```

```
    printf("\n--- Inventory Details ---\n");

    for(i = 0; i < n; i++) {
        printf("%d  %s  %d  %.2f\n",
               item[i].id,
               item[i].name,
               item[i].quantity,
               item[i].price);
    }

    return 0;
}
```

OUTPUT:

```
Enter number of items: 3

Enter item ID: 12
Enter item name: pen
Enter quantity: 4
Enter price: 45

Enter item ID: 23
Enter item name: pencil
Enter quantity: 6
Enter price: 30

Enter item ID: 67
Enter item name: scale
Enter quantity: 8
Enter price: 80

--- Inventory Details ---
Id=12 Name=pen Quantity=4 price=45.00
Id=23 Name=pencil Quantity=6 price=30.00
Id=67 Name=scale Quantity=8 price=80.00
```

7.Decimal to binary, octal and hexadecimal

```
#include <stdio.h>

int main()
{
    int n, temp, rem;
    int binary[50], octal[50], hex[50];
    int i, j, k;

    printf("Enter decimal number: ");
    scanf("%d", &n);

    /* Binary */
```

```
temp = n;
i = 0;

while(temp > 0)
{
    rem = temp % 2;
    binary[i] = rem;
    i++;
    temp = temp / 2;
}

printf("Binary = ");

for(i = i - 1; i >= 0; i--)
{
    printf("%d", binary[i]);
}

/* Octal */
temp = n;
j = 0;

while(temp > 0)
{
    rem = temp % 8;
    octal[j] = rem;
    j++;
    temp = temp / 8;
}

printf("\nOctal = ");

for(j = j - 1; j >= 0; j--)
{
    printf("%d", octal[j]);
}

/* Hexadecimal */
temp = n;
k = 0;

while(temp > 0)
{
    rem = temp % 16;
    hex[k] = rem;
    k++;
    temp = temp / 16;
}

printf("\nHexadecimal = ");
```

```
    for(k = k - 1; k >= 0; k--)
    {
        if(hex[k] < 10)
            printf("%d", hex[k]);
        else
            printf("%c", hex[k] - 10 + 'A');
    }

    return 0;
}
```

```
OUTPUT:
Enter decimal number: 56
Binary = 111000
Octal = 70
Hexadecimal = 38
```

8.diffference between two time periods

```
#include <stdio.h>

struct Time {
    int hour;
    int minute;
    int second;
};

int main() {
    struct Time t1, t2, diff;

    printf("Enter first time (hh mm ss): ");
    scanf("%d %d %d", &t1.hour, &t1.minute, &t1.second);

    printf("Enter second time (hh mm ss): ");
    scanf("%d %d %d", &t2.hour, &t2.minute, &t2.second);

    if(t2.second < t1.second) {
        t2.second += 60;
        t2.minute--;
    }

    if(t2.minute < t1.minute) {
        t2.minute += 60;
        t2.hour--;
    }

    diff.second = t2.second - t1.second;
    diff.minute = t2.minute - t1.minute;
    diff.hour = t2.hour - t1.hour;

    printf("Difference = %d:%d:%d",
```

```
           diff.hour, diff.minute, diff.second);

    return 0;
}
OUTPUT:
Enter first time (hh mm ss): 06 34 25
Enter second time (hh mm ss): 12 56
23
Difference = 6:21:58
--------------------------------------------------------------------------------
9.Employee management system using structure
#include <stdio.h>

struct Employee {
    int id;
    char name[50];
    float salary;
};

int main() {
    struct Employee e[50];
    int n, i;

    printf("Enter number of employees: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("\nEnter employee ID: ");
        scanf("%d", &e[i].id);

        printf("Enter name: ");
        scanf("%s", e[i].name);

        printf("Enter salary: ");
        scanf("%f", &e[i].salary);
    }

    printf("\n--- Employee Details ---\n");

    for(i = 0; i < n; i++) {
        printf("ID: %d\n", e[i].id);
        printf("Name: %s\n", e[i].name);
        printf("Salary: %.2f\n\n", e[i].salary);
    }

    return 0;
}
OUTPUT:
Enter number of employees: 4
```

```
Enter employee ID: 115
Enter name: pavatharani
Enter salary: 300000

Enter employee ID: 116
Enter name: gobika
Enter salary: 450000

Enter employee ID: 117
Enter name: arun
Enter salary: 500000

Enter employee ID: 120
Enter name: sathiya
Enter salary: 250000

--- Employee Details ---
ID: 115
Name: pavatharani
Salary: 300000.00

ID: 116
Name: gobika
Salary: 450000.00

ID: 117
Name: arun
Salary: 500000.00

ID: 120
Name: sathiya
Salary: 250000.00
```

10.Area and perimeter of rectangle and triangle

```
#include <stdio.h>

void rectangle(float *l, float *b) {
    printf("Rectangle Area = %.2f\n", (*l) * (*b));
    printf("Rectangle Perimeter = %.2f\n",
           2 * ((*l) + (*b)));
}

void triangle(float *a, float *b, float *c) {
    float s, area;

    s = (*a + *b + *c) / 2;

    area = (s * (s - *a) * (s - *b) * (s - *c));

    printf("Triangle Perimeter = %.2f\n",
```

```
           *a + *b + *c);

    printf("Triangle Area = %.2f\n",
           area);
}

int main() {
    float l, b;
    float a, c, d;

    printf("Enter length and breadth of rectangle: ");
    scanf("%f %f", &l, &b);

    rectangle(&l, &b);

    printf("\nEnter three sides of triangle: ");
    scanf("%f %f %f", &a, &c, &d);

    triangle(&a, &c, &d);

    return 0;
}
OUTPUT:
Enter length and breadth of rectangle: 12
34
Rectangle Area = 408.00
Rectangle Perimeter = 92.00

Enter three sides of triangle: 4 6 7
Triangle Perimeter = 17.00
Triangle Area = 143.44
--------------------------------------------------------------------------------
```